

Simulation Results

Smart-Root-Dumb-Tree Model

Search scope : Combat move selection only - attacks only 1 territory in a round

Selection policy : UCB1 with exploration constant $c = 1.414$

Expansion : Single node per iteration

Simulation policy : Uniform random rollout for all players - keeps the tree dumb and root smart - excludes purchase and placement delegates - the simulation move generation has no pruning apart from time budget limit

Forward model : Handles the logs from the game, provides the requested move for the MCTS player (tree selects for combat and random for other delegates)

State evaluation : Territory strength ration + depth based penalties for non-terminal states

Playout : Fast playout by speeing legal move genration using game heuristics. The combat move generator gives 1 move with cheapest unit for undefended territory and a set of winnable moves with [min, max] strength for defended territories

Time budget : 1s per combat turn

Max tree depth : 10

Goal : Isolate and measure combat decision quality independent of the other game delegates

The legal move generation is perfomed in Cpp and the linked to the python script using Cython

Efficiency summary

MCTS Efficiency Summary (Game-wise)

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Game 1

Total rounds: 31

Iterations per round:

Mean: 40.71

Min : 30

Max : 107

Exploration_coverage (actions_explored / total_actions):

Mean: 0.3341

Min : 0.0003

Max : 1.0000

Avg Tree depth per round:

Mean: 9.6353

Min : 2.5888

Max : 10.0000

Game 2

Total rounds: 22

Iterations per round:

Mean: 39.18

Min : 33

Max : 71

Exploration_coverage (actions_explored / total_actions):

Mean: 0.3807

Min : 0.0015

Max : 1.0000

Avg Tree depth per round:

Mean: 9.9194

Min : 9.7353

Max : 10.0000

Game 3

Total rounds: 31

Iterations per round:

Mean: 33.23

Min : 28

Max : 71

Exploration_coverage (actions_explored / total_actions):

Mean: 0.3935

Min : 0.0003

Max : 1.0000

Avg Tree depth per round:

Mean: 9.8994

Min : 9.7188

Max : 10.0000

Game 4

Total rounds: 66

Iterations per round:

Mean: 36.51

Min : 27

Max : 60

Exploration_coverage (actions_explored / total_actions):

Mean: 0.1846

Min : 0.0000

Max : 1.0000

Avg Tree depth per round:

Mean: 9.8920

Min : 9.7097

Max : 10.0000

Game 5

Total rounds: 25

Iterations per round:

Mean: 38.80

Min : 34

Max : 59

Exploration_coverage (actions_explored / total_actions):

Mean: 0.5557

Min : 0.0061

Max : 1.0000

Avg Tree depth per round:

Mean: 9.8776

Min : 9.7429

Max : 10.0000

OVERALL AVERAGES (Across Games)

Iterations per round (game-averaged):

Mean: 37.68

Min: 27

Max: 107

Exploration coverage (game-averaged):

Mean: 0.37

Min: 0.00

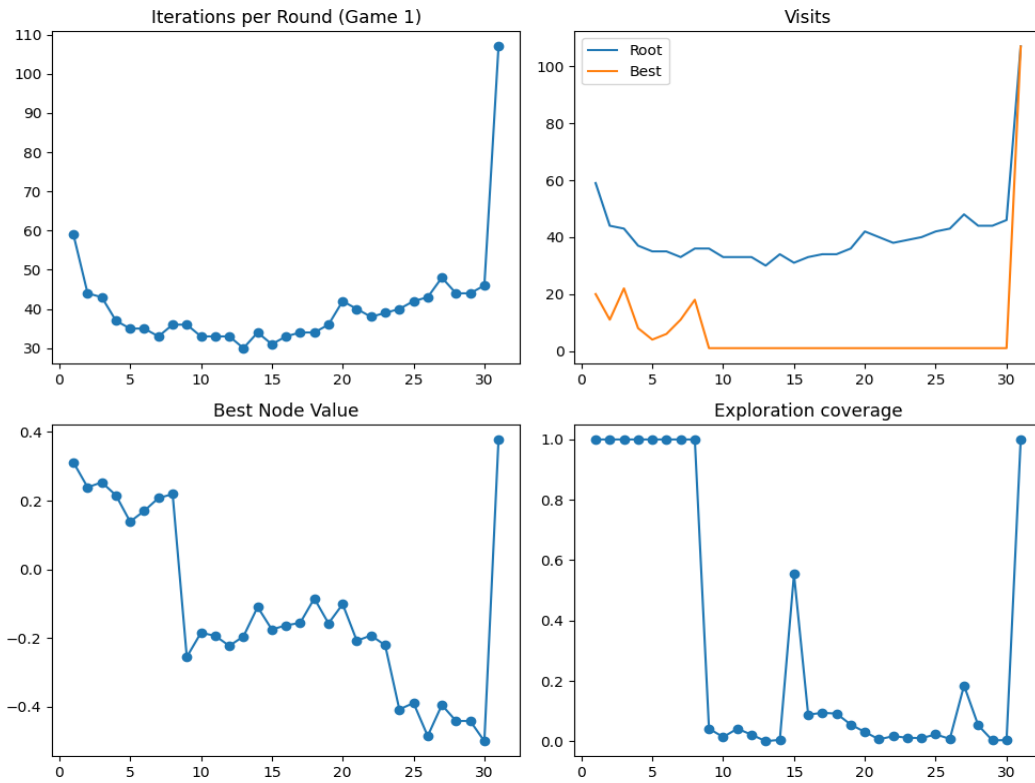
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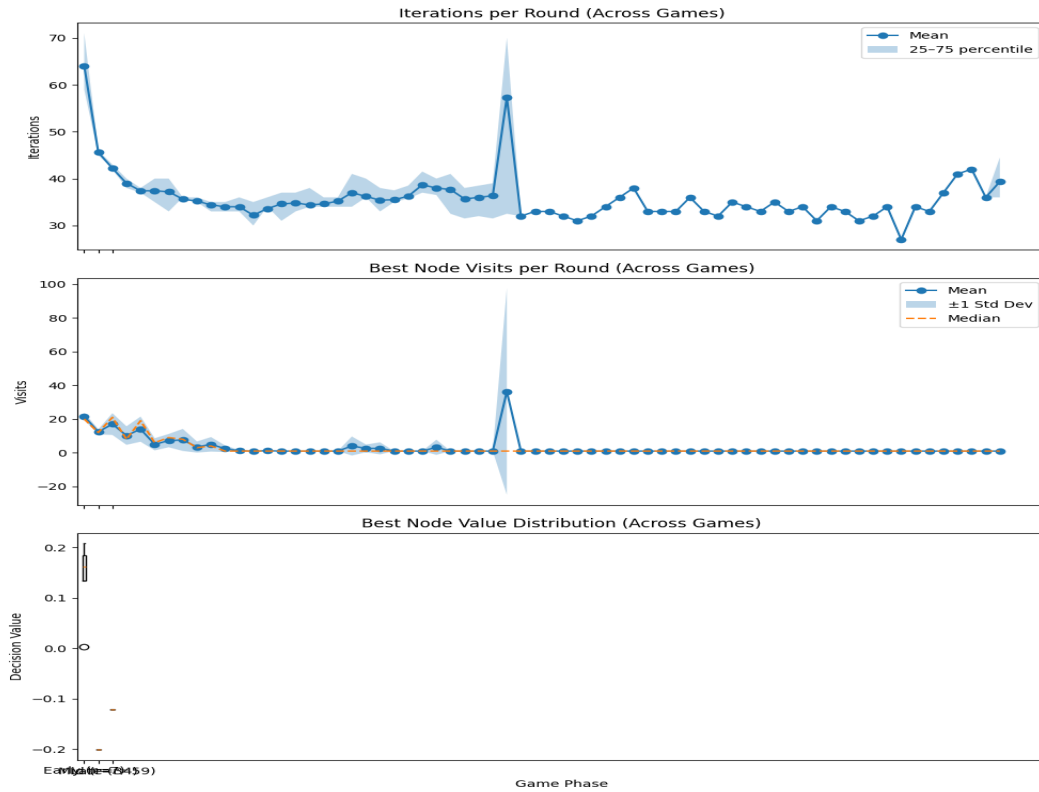
Avg tree depth per round (game-averaged):

Mean: 9.84

Min: 2.588785046728972

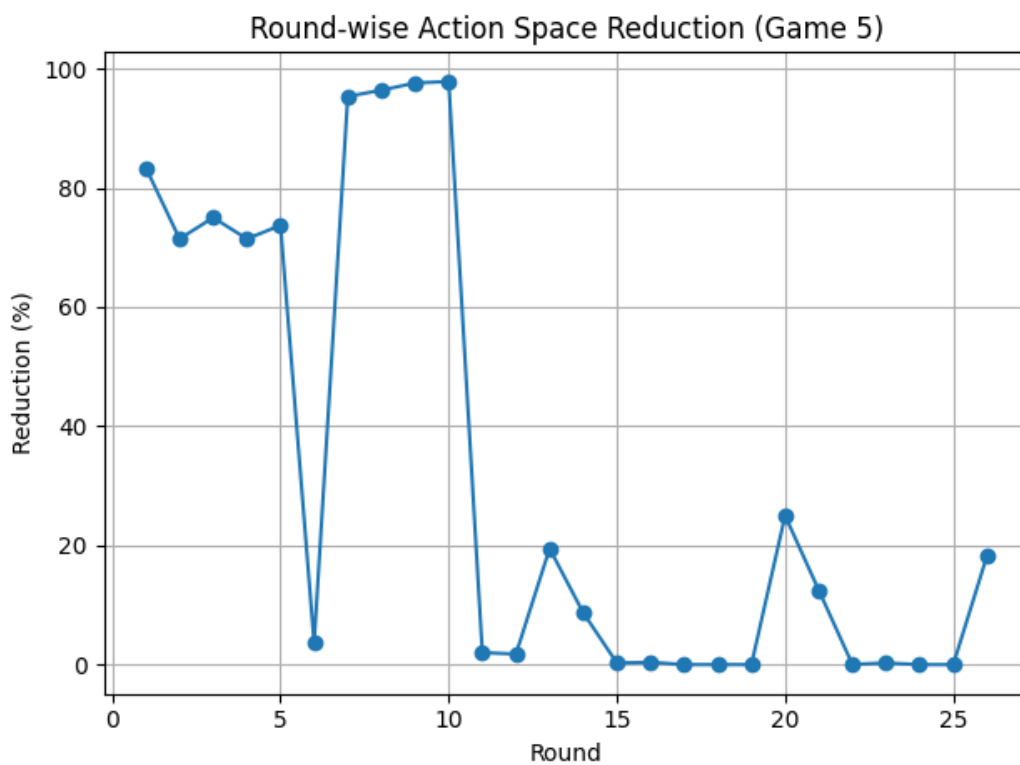
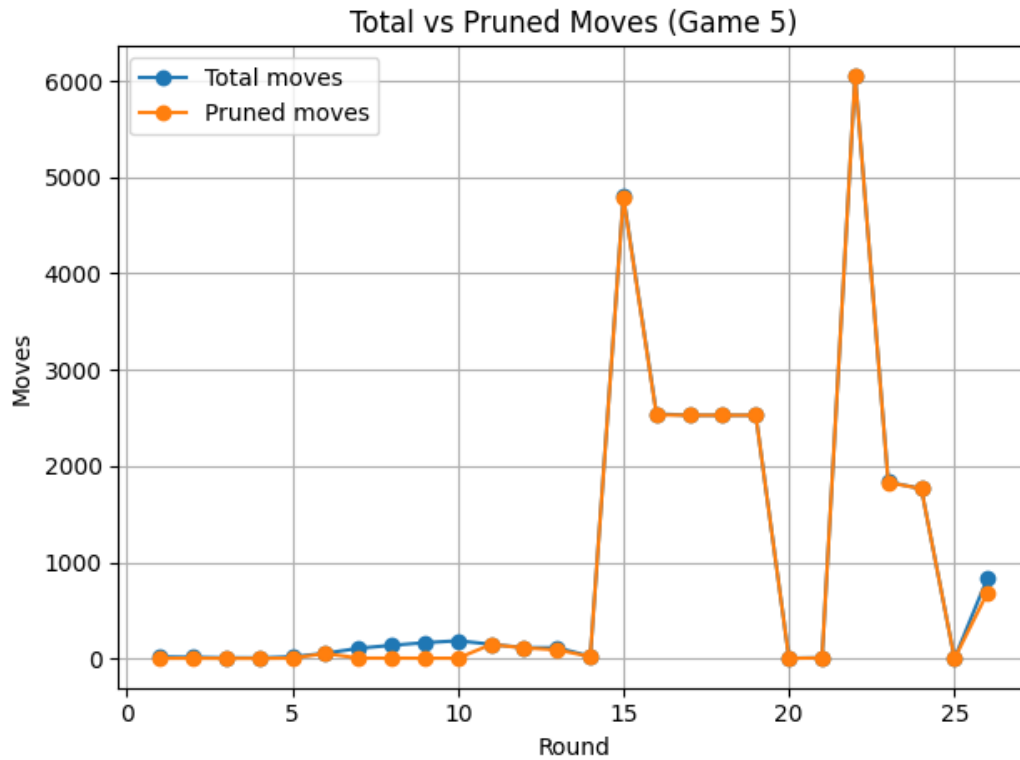
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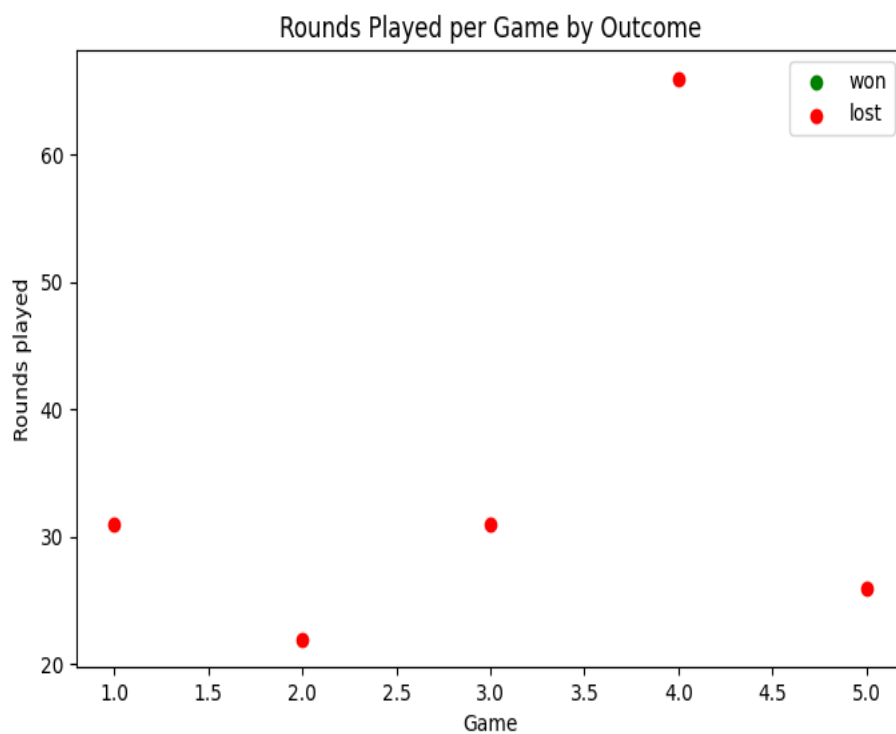
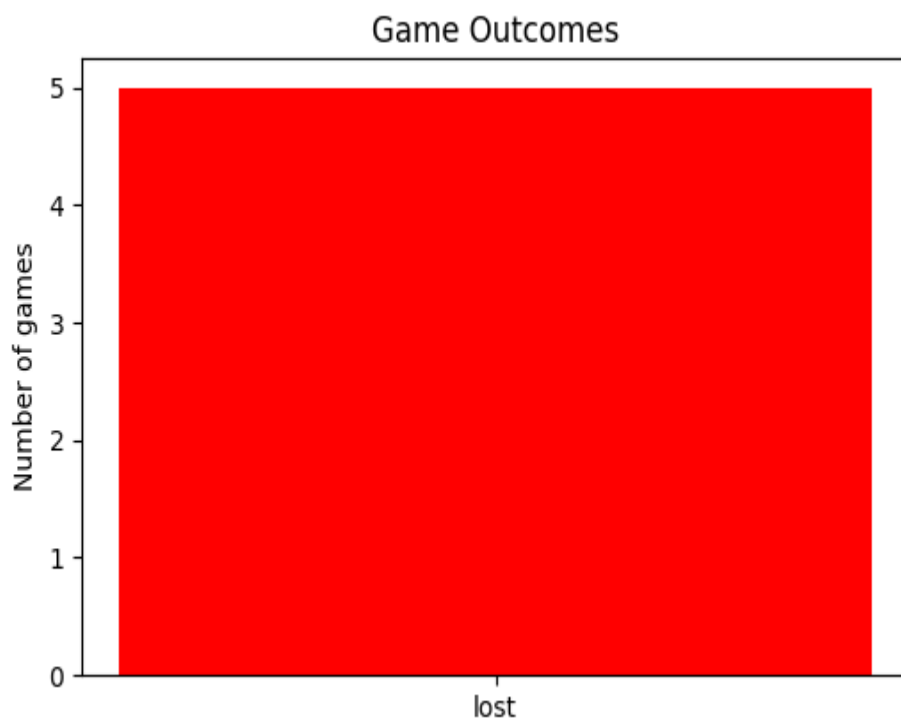


Action space reduction summary

game	rounds	avg_round_reduction	avg_round_reduction_percent
1	31.0	0.311592	31.159197
2	22.0	0.310293	31.029325
3	31.0	0.324613	32.461322
4	87.0	0.202694	20.269387
5	26.0	0.328562	32.856184
OVERALL	39.4	0.295551	29.555083



Game outcomes

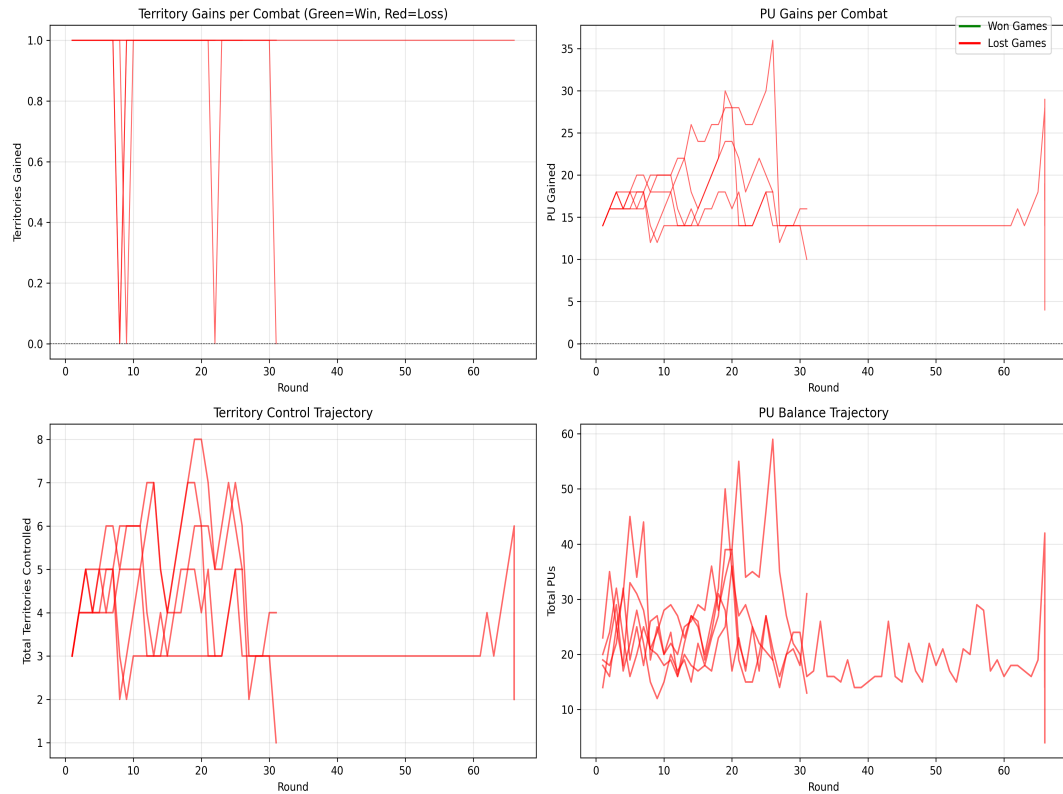


Combat move quality throughout the games

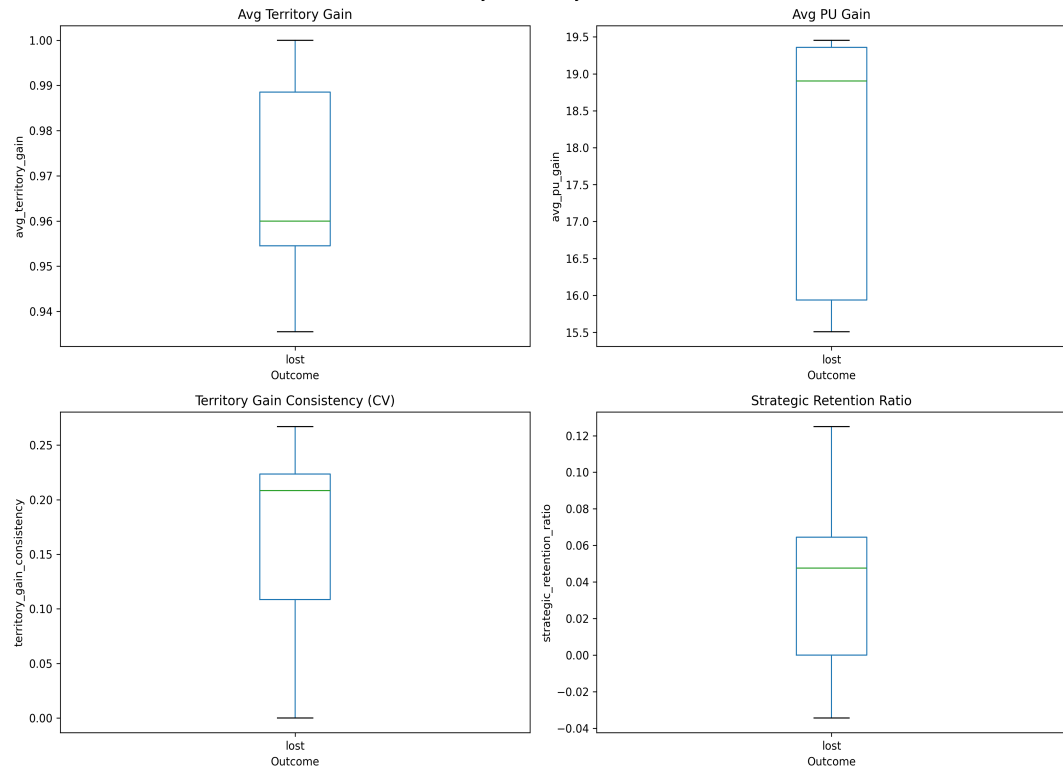
=== Combat Quality Metric Averages ===

Outcome: LOST

Avg Territory Gain: 0.968
Avg PU Gain: 17.832
Territory Gain Consistency: 0.161
Strategic Retention Ratio: 0.041



Combat Quality Metrics by Game Outcome



Profiling of MCTS algorithm

